



9482 - Time-Critical Follow-Up of Newly Discovered Uranian Moon with JWST/NIRCam

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Uranus	Deep Imaging			
	1		NIRCam Imaging	(1) Uranus
	2	Residual Persistence Check	NIRCam Dark	NONE

ABSTRACT

We request time-critical Director’s Discretionary Time (DDT) on JWST to reobserve S/2025 U 1, a newly discovered inner satellite of Uranus. This object was identified in General Observer Program #6379 on 2025 February 2 UT, from a sequence of ten long-exposure NIRCam images. The satellite orbits between Uranus VII (Ophelia) and Uranus VIII (Bianca), at a projected distance of 56,250 +/- 250 km from Uranus’ center. Its mean

motion of 896 ± 1 degrees/day corresponds to a 9.65-hour orbital period. This discovery raises Uranus' satellite count to 29 and opens a rare, new window into the dynamics of the planet's poorly understood and densely-packed inner ring-moon system. However, the object's short orbital period and close proximity to Uranus severely constrain the observational timeline. Without a second-epoch observation within 7–9 months of the discovery, the satellite's ephemeris uncertainty will grow to the point that confirmation and orbit prediction become much more difficult.

OBSERVING DESCRIPTION

We will use NIRCcam to obtain our deepest survey of this newly discovered satellite in the Uranus system. These images will be centered on Uranus and employ the FULLP subarray, providing a 900,000-km field of view in the short-wavelength detector. We will use the two widest filters, F150W2 and F322W2, to maximize our sensitivity to this small moon. Although Uranus will be heavily saturated, the moons will not be. Similar observations of the Uranus system in GO program 6379 have shown that NIRCcam is quite capable of obtaining clear images of faint moons adjacent to a heavily overexposed planet.

Our key goal for these observations are to re-find this small moon in the system and to characterize its orbit. We propose a brief but critically timed JWST/NIRCcam observation to reobserve and secure the orbit of S/2025 U 1, a faint, newly discovered inner satellite of Uranus. This follow-up, scheduled ~ 240 days after the initial detection, will reduce orbital uncertainties by three orders of magnitude, enabling dynamic modeling and future observability. Additional candidate moons will also be verified or ruled out. This DDT request enables high-impact science with minimal time investment.

Proposal 9482 - Targets - Time-Critical Follow-Up of Newly Discovered Uranian Moon with JWST/NIRCam

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	Uranus	STD=URANUS		
	Comments: Extended=YES				

Proposal 9482 - Observation 1 - Time-Critical Follow-Up of Newly Discovered Uranian Moon with JWST/NIRCam

Observation	Proposal 9482, Observation 1 Diagnostic Status: Warning Observing Template: NIRCam Imaging										Thu Sep 18 20:00:32 GMT 2025	
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Observation 1) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Diagnostics												
Solar System Targets	#	Name	Level 1			Level 2			Level 3			
	(1)	Uranus	STD=URANUS									
Template	Module B											Subarray FULLP
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	NONE				SMALL-GRID-DITHER				6		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F150W2	F322W2	BRIGHT1	10	10	60	6	12819.703			
Special Requirements	Offset -11.0 arcsec, -11.0 arcsec Persistence Risk											
	2 After 1 by 8.5 Hours to 2 Days, Exclusive Use Of Instrument DEFAULT WINDOW: ANGULAR RATE URANUS FROM JWST LESS THAN 0.075											

Proposal 9482 - Observation 2 - Time-Critical Follow-Up of Newly Discovered Uranian Moon with JWST/NIRCam

Observation	Proposal 9482, Observation 2: Residual Persistence Check								Thu Sep 18 20:00:32 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRCam Dark								
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Template	Module	Subarray		Science Template		Occulting Mask		No. of Output Channels	
	B	FULL		Imaging				4	
Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	MEDIUM8	10	1	1	1	1052.203		
Special Requirements	2 After 1 by 8.5 Hours to 2 Days, Exclusive Use Of Instrument								